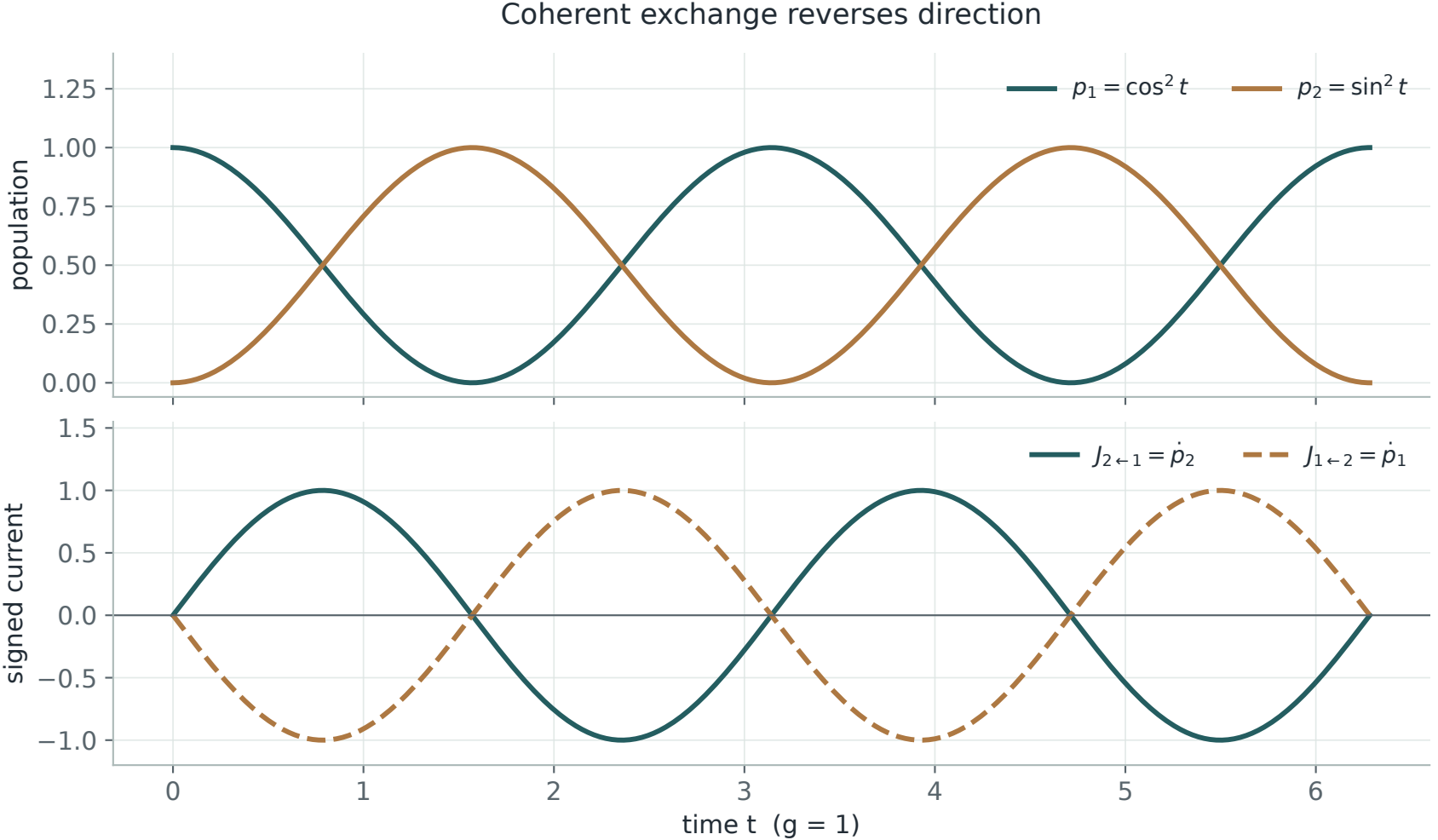
Coherent exchange reverses direction
population
1.25
1.00
0.75
0.50
0.25
0.00
$p_1 = \cos^2 t$
$p_2 = \sin^2 t$
signed current
1.5
1.0
0.5
0.0
−0.5
−1.0
$J_{2\leftarrow 1} = \dot{p}_2$
$J_{1\leftarrow 2} = \dot{p}_1$
0
1
2
3
4
5
6
time t (g = 1)